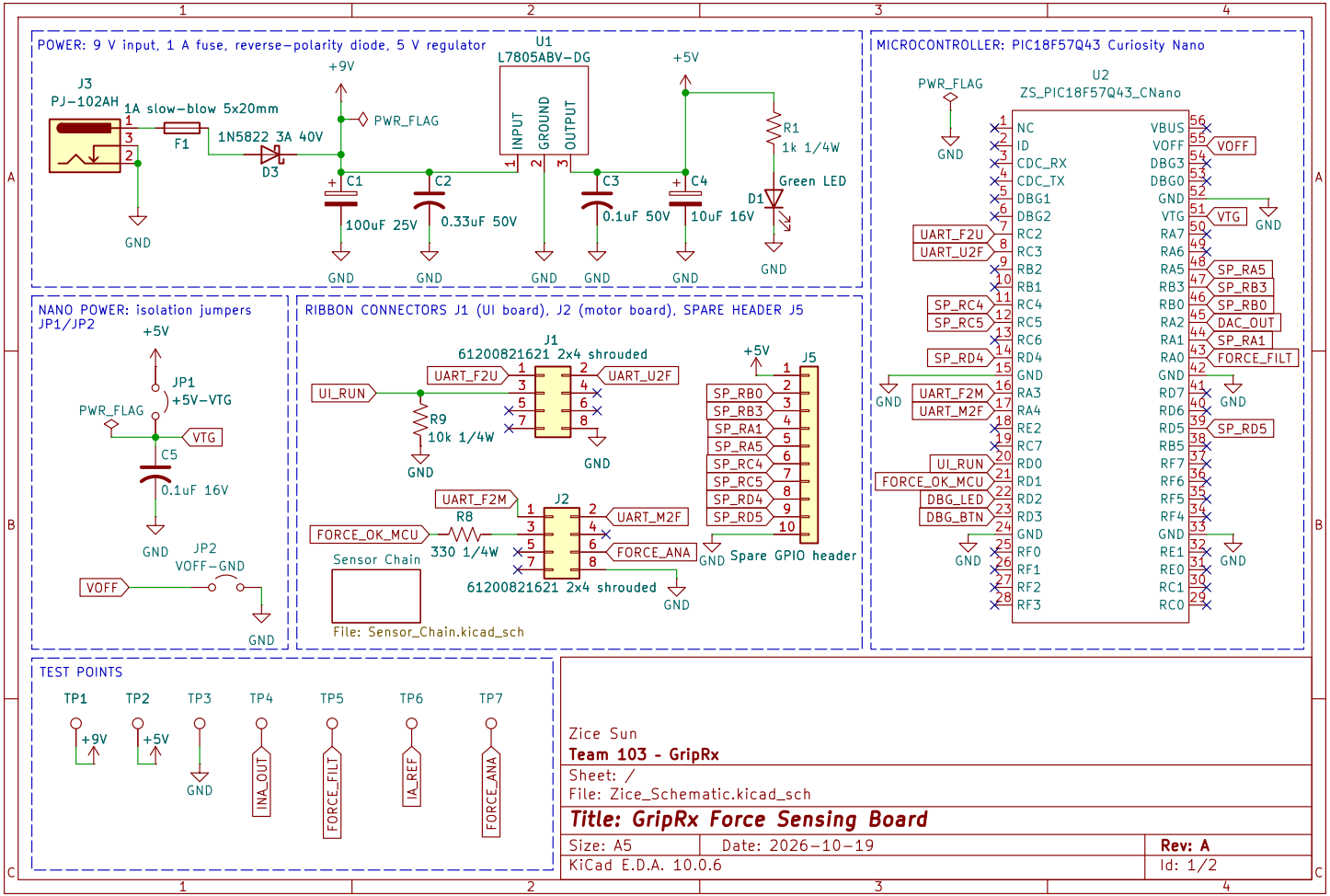
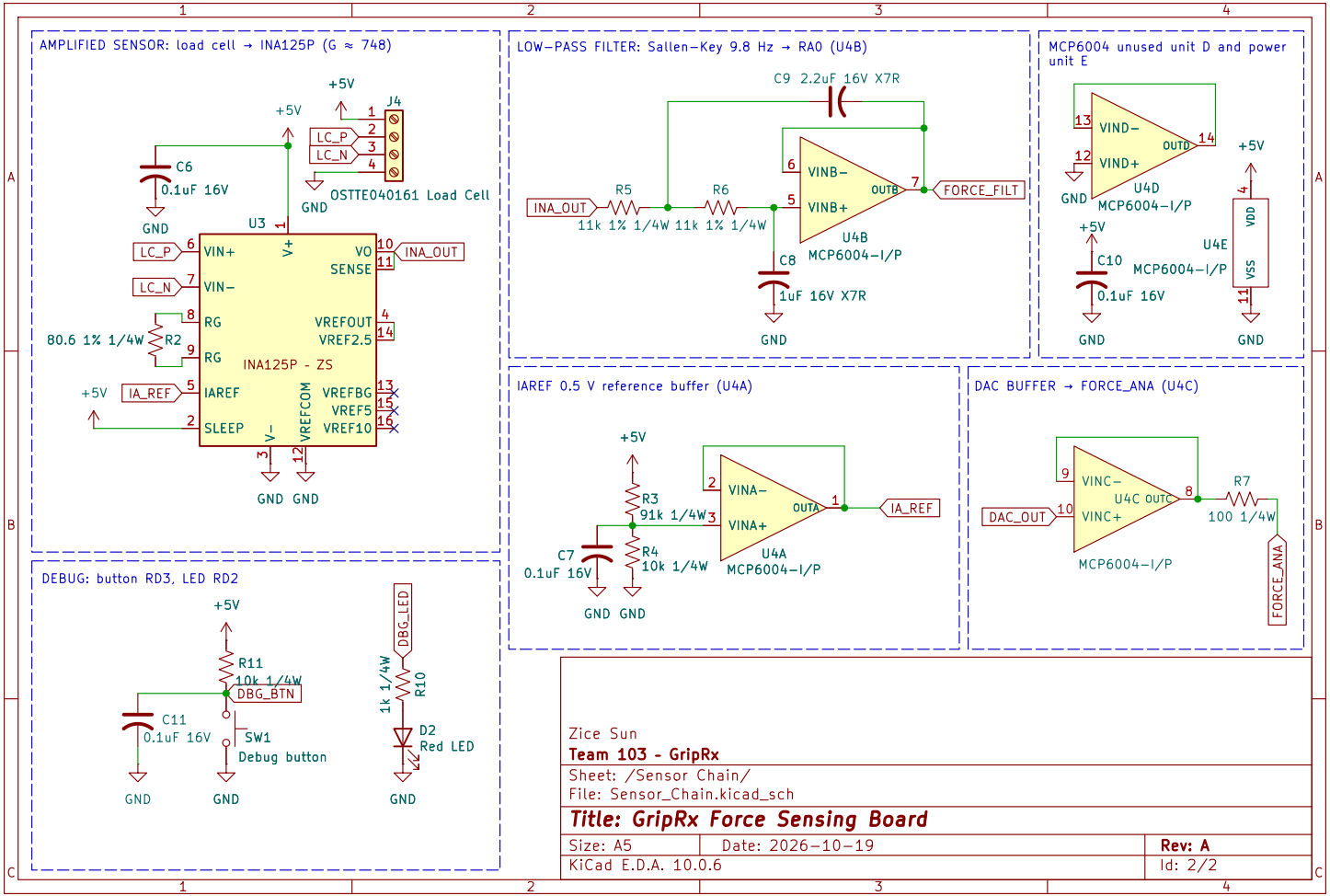

POWER: 9 V input, 1 A fuse, reverse-polarity diode, 5 V regulator
J3
PJ-102AH
1A slow-blow 5x20mm
F1
1N5822 3A 40V
D3
+9V
PWR_FLAG
GND
U1
L7805ABV-DG
INPUT
GROUND
OUTPUT
C1
100uF 25V
C2
0.33uF 50V
C3
0.1uF 50V
C4
10uF 16V
+5V
R1
1k 1/4W
D1
Green LED
MICROCONTROLLER: PIC18F57Q43 Curiosity Nano
U2
ZS_PIC18F57Q43_CNano
NANO POWER: isolation jumpers
JP1/JP2
JP1
+5V-VTG
VTG
C5
0.1uF 16V
JP2
VOFF-GND
VOFF
RIBBON CONNECTORS J1 (UI board), J2 (motor board), SPARE HEADER J5
J1
61200821621 2x4 shrouded
UART_F2U
UART_U2F
UI_RUN
R9
10k 1/4W
J2
UART_F2M
UART_M2F
FORCE_OK_MCU
R8
330 1/4W
FORCE_ANA
Sensor Chain
File: Sensor_Chain.kicad_sch
J5
SP_RB0
SP_RB3
SP_RA1
SP_RA5
SP_RC4
SP_RC5
SP_RD4
SP_RD5
Spare GPIO header
TEST POINTS
TP1
TP2
TP3
TP4
TP5
TP6
TP7
INA_OUT
FORCE_FILT
IA_REF
FORCE_ANA
Zice Sun
Team 103 - GripRx
Sheet: /
File: Zice_Schematic.kicad_sch
Title: GripRx Force Sensing Board
Size: A5
Date: 2026-10-19
Rev: A
KiCad E.D.A. 10.0.6
Id: 1/2


AMPLIFIED SENSOR: load cell → INA125P (G ≈ 748)
+5V
J4
LC_P
LC_N
OSTTE040161 Load Cell
GND
C6
0.1uF 16V
U3
INA125P - ZS
VIN+
VIN−
RG
RG
IAREF
SLEEP
V+
V−
VREFCOM
VO
SENSE
VREFOUT
VREF2.5
VREFBG
VREF5
VREF10
INA_OUT
80.6 1% 1/4W
R2
IA_REF
GND GND
LOW−PASS FILTER: Sallen−Key 9.8 Hz → RA0 (U4B)
C9 2.2uF 16V X7R
R5
R6
11k 1% 1/4W 11k 1% 1/4W
VINB−
VINB+
OUTB
U4B
MCP6004−I/P
FORCE_FILT
C8
1uF 16V X7R
GND
MCP6004 unused unit D and power unit E
VIND−
VIND+
OUTD
U4D
MCP6004−I/P
U4E
VDD
VSS
C10
0.1uF 16V
IAREF 0.5 V reference buffer (U4A)
R3
91k 1/4W
R4
10k 1/4W
C7
0.1uF 16V
VINA−
VINA+
OUTA
U4A
MCP6004−I/P
DAC BUFFER → FORCE_ANA (U4C)
DAC_OUT
VINC−
VINC+
U4C OUTC
R7
100 1/4W
FORCE_ANA
DEBUG: button RD3, LED RD2
R11
10k 1/4W
DBG_BTN
C11
0.1uF 16V
SW1
Debug button
DBG_LED
R10
1k 1/4W
D2
Red LED
Zice Sun
Team 103 - GripRx
Sheet: /Sensor Chain/
File: Sensor_Chain.kicad_sch
Title: GripRx Force Sensing Board
Size: A5
Date: 2026−10−19
Rev: A
KiCad E.D.A. 10.0.6
Id: 2/2